



# Darshan University

A Project Report on

**“Website Archiver”**

Under the subject

**Software Engineering (2301CS405)**

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Computer Science & Engineering Department

Submitted By

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Internal Guide

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**DECLARATION**

We hereby declare that the SRS, submitted along with the **Software Engineering (2301CS405)** for entitled “**Website Archiver**” submitted in partial fulfilment for the Semester-5 of **Bachelor Technology (B. Tech)** in **Computer Science and Engineering (CSE)** Department to Darshan University, Rajkot, is a record of the work carried out at **Darshan University, Rajkot** under the supervision of R. B. Gondaliya and that no part of any of report has been directly copied from any students’ reports, without providing due reference.

**Sagathiya Soham**

Student’s Signature

Date: 29/12/2024



**Computer Science & Engineering  
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**CERTIFICATE**

This is to certify that the SRS on “**Website Archiver**” has been satisfactorily prepared by **Sagathiya Soham (24010101682)** under my guidance in the fulfillment of the course **Software Engineering (2301CS405)** work during the academic year 2024-2025.

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Thanking You

**Sagathiya Soham**

## ABSTRACT

Website Archiver is a platform designed to periodically archive static websites and make them accessible for browsing. The system allows users to view, compare, and download archived versions of websites, with different access levels for guests, viewers, and Members. Guests can browse public archives, search for websites by date or name, and view them in a calendar format. Viewers, after registering and logging in, can access private archives, generate public links, and download archives in ZIP format. Members have additional privileges to manage the websites being archived, set archiving schedules, modify visibility, and receive notifications for failed tasks or storage limits.

The system ensures efficient archiving of static website content, including HTML, CSS, JS, and images, with the ability to handle dynamic content through basic JavaScript rendering. It offers scalable storage options and supports error handling, retries, and analytics tracking. With a focus on performance, reliability, and security, the platform provides a user-friendly interface for both website owners and visitors.

The main goal of the Website Archiver is to streamline the process of archiving websites, reducing manual effort and enabling easy access to past versions for various use cases such as content retrieval, historical reference, and legal compliance.

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# 1 Introduction

## 1.1 Product perspective

This project aims to transition the manual process of archiving websites into an automated internet-based application, allowing users to access, browse, and manage archived content conveniently. The system ensures the periodic capture of website content, making it available in a browsable format while catering to multiple user roles, including guests, viewers, and Members. It offers core functionalities like website archiving, scheduling, metadata management, and accessibility control. Designed as a multi-user system, it can efficiently handle the fundamental archiving needs for individuals, organizations, or small enterprises.

## 1.2 Product features

1.2.1 The system provides features based on three distinct roles:

- **Guest:** Non-authorized users who can access public archives.
- **Viewer:** Authorized users with enhanced access to private archives and additional tools.
- **Member:** Authorized users with full control over archiving settings and website management.

1.2.2 Guest Features:

- View public archived websites.
- Browse archived versions by date or website.
- Search archives using website names or specific dates.
- Access archives via a calendar-based view.

1.2.3 Viewer Features (includes Guest capabilities):

- Register and log in to the platform.
- View private archived websites (with granted access).
- Compare different versions of an archive side-by-side.
- Generate public links to share archives.
- Download archived websites as compressed ZIP files.

1.2.4 Member Features (includes Viewer capabilities):

- Add websites for periodic archiving.
- Remove websites from the archiving list.
- Update or modify archiving schedules.
- Delete outdated archives they own.
- Adjust archive visibility between public and private.
- Receive notifications about failed archiving tasks or storage nearing capacity.
- Monitor storage usage statistics.
- Export and import archive metadata in JSON format.

## 1.3 Functional Requirement

1.3.1 System-Wide Functionalities

- Periodically archive static website content, including HTML, CSS, JS, and images.
- Handle basic JavaScript rendering to support dynamic content.
- Compress and store archived files to save space.
- Log and retry failed archiving tasks.
- Track analytics, such as archive counts, storage consumption, and page views.

1.3.2 Member Functionalities

- **Add Websites:** Allow Members to include websites for archiving.
- **Update Schedule:** Modify archiving intervals (daily, weekly, monthly, yearly).

- **Delete Archives:** Remove specific archives when no longer required.
- **Set Permissions:** Control access to archives (public or private).
- **View Statistics:** Monitor storage usage and task completion rates.

#### 1.3.3 Viewer Functionalities

- **Authentication:** Authenticate users for private archive access.
- **Compare Archives:** View two versions of an archive side-by-side.
- **Public Links:** Share archives via public links.
- **Download Archives:** Download archived websites as ZIP files.

#### 1.3.4 Guest Functionalities

- **Browse Archives:** View public archives and browse them by date or name.
- **Search Archives:** Use search functionality to find specific archives.

### 1.4 Non-Functional Requirement

#### 1.4.1 Usability:

- Provide a simple, intuitive user interface for all user roles.
- Offer multilingual support as needed.
- Ensure responsive design for compatibility with desktop and mobile devices.

#### 1.4.2 Accuracy:

- Guarantee accurate archiving of website content, ensuring all assets are captured.

#### 1.4.3 Availability:

- Ensure the platform is operational 24/7 with downtime recovery within one hour.

#### 1.4.4 Maintainability:

- Maintain a modular and easily extensible codebase for adding new features.

#### 1.4.5 Scalability:

- Allow for concurrent archiving of multiple websites without performance degradation.
- Scale storage systems to accommodate increasing archive volumes.

## 2 Design and Implementation Constraints

### 2.1 Use case diagram

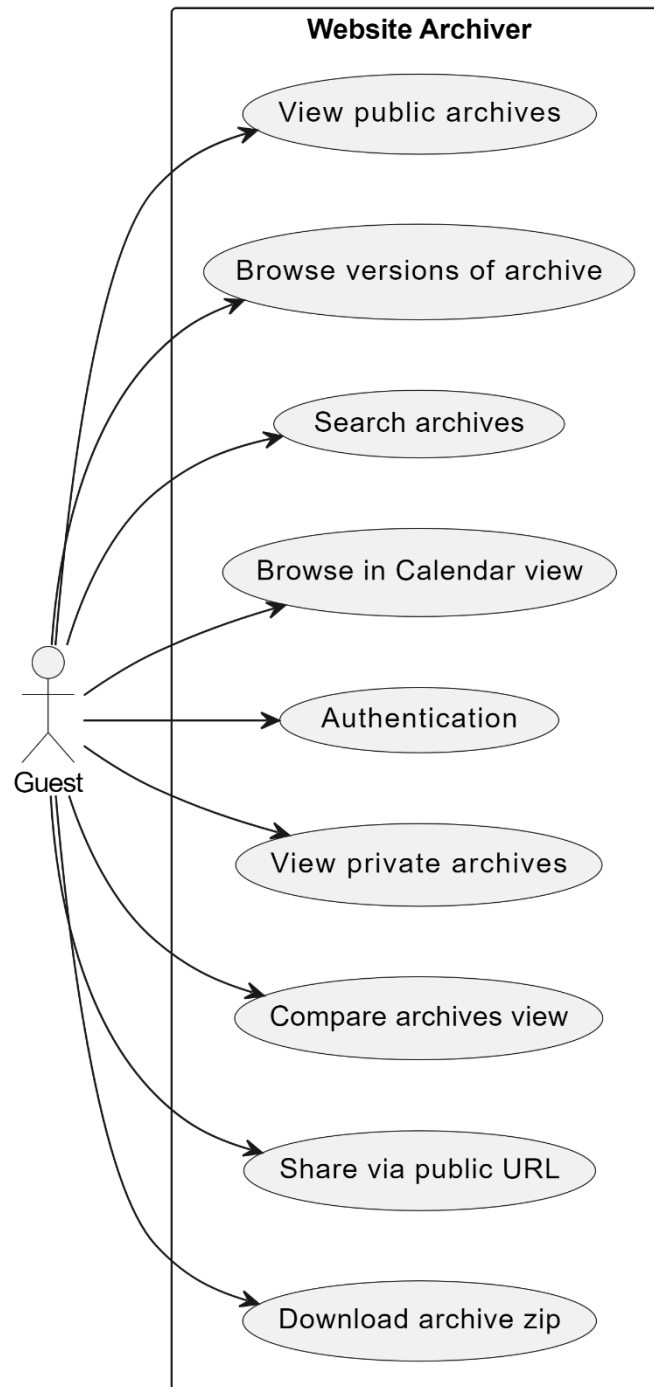


Figure 2.1-1 Use case diagram for Web Archiver (Guest / Viewer)

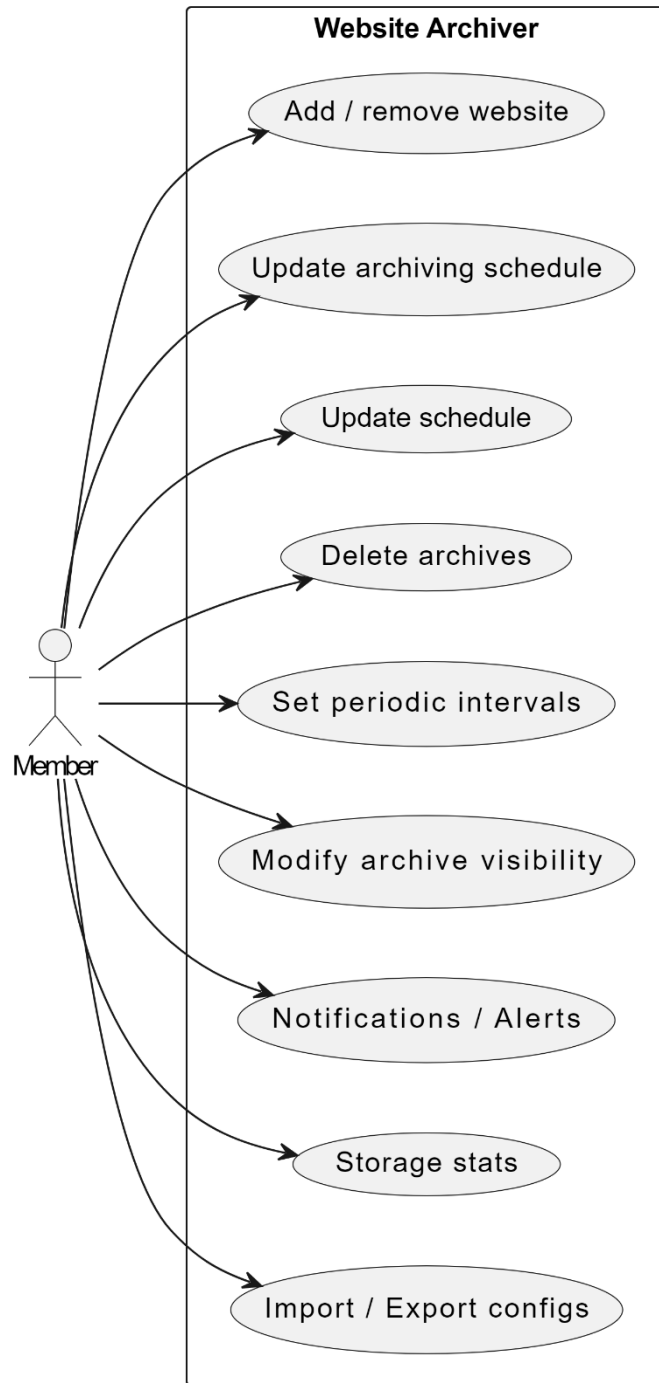


Figure 2.1-2 Use case diagram for Web Archiver (Member)

## 2.2 Activity diagram and Swimlane diagram

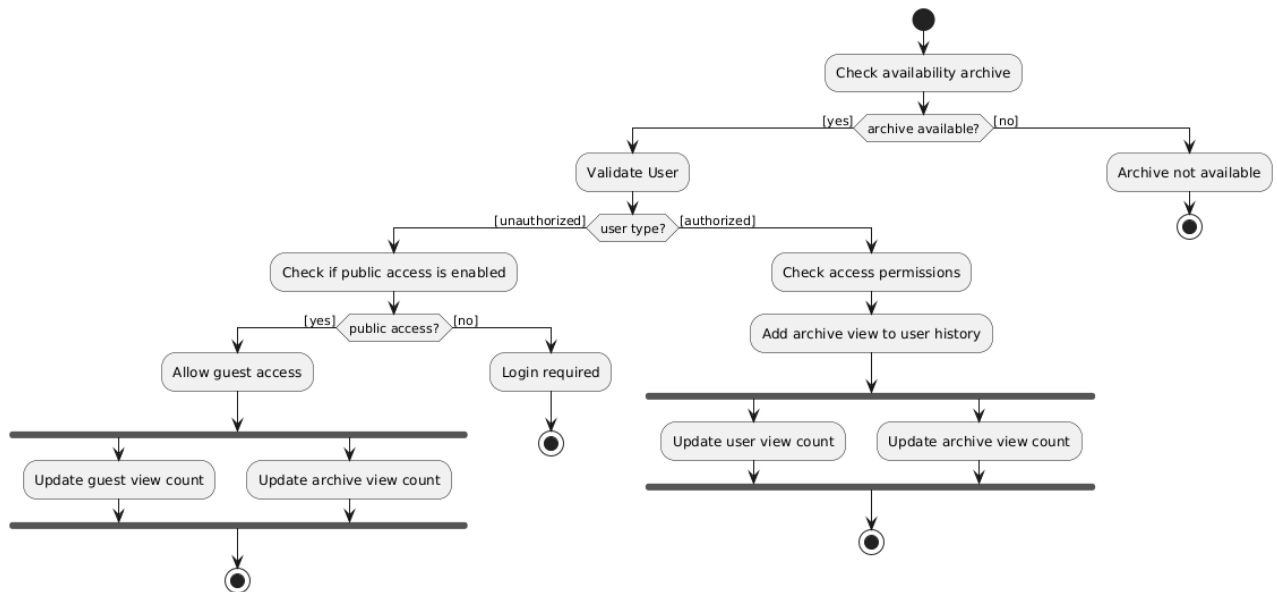


Figure 2.2-1 Activity diagram for viewing archived site

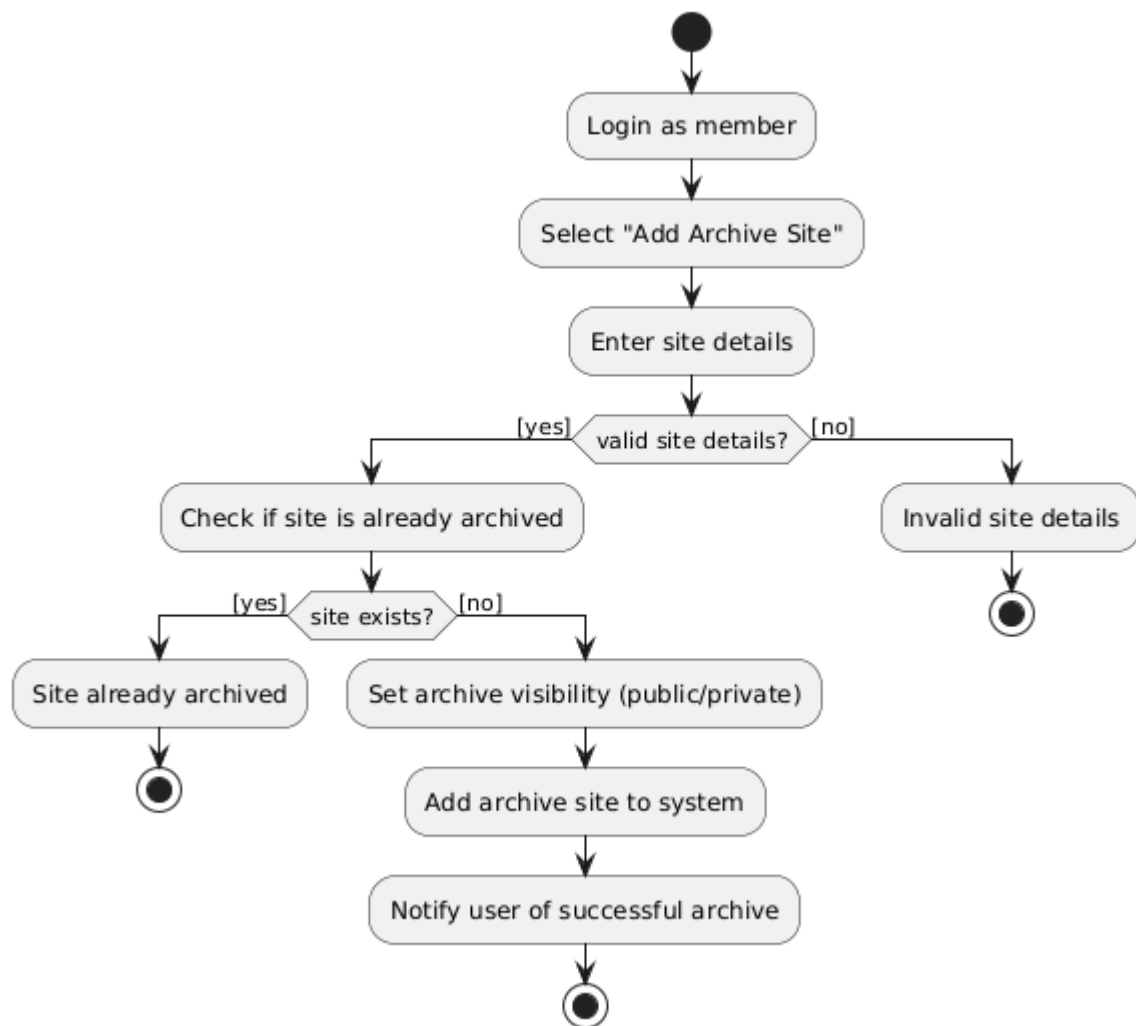


Figure 2.2-2 Activity diagram for adding a site for archiving

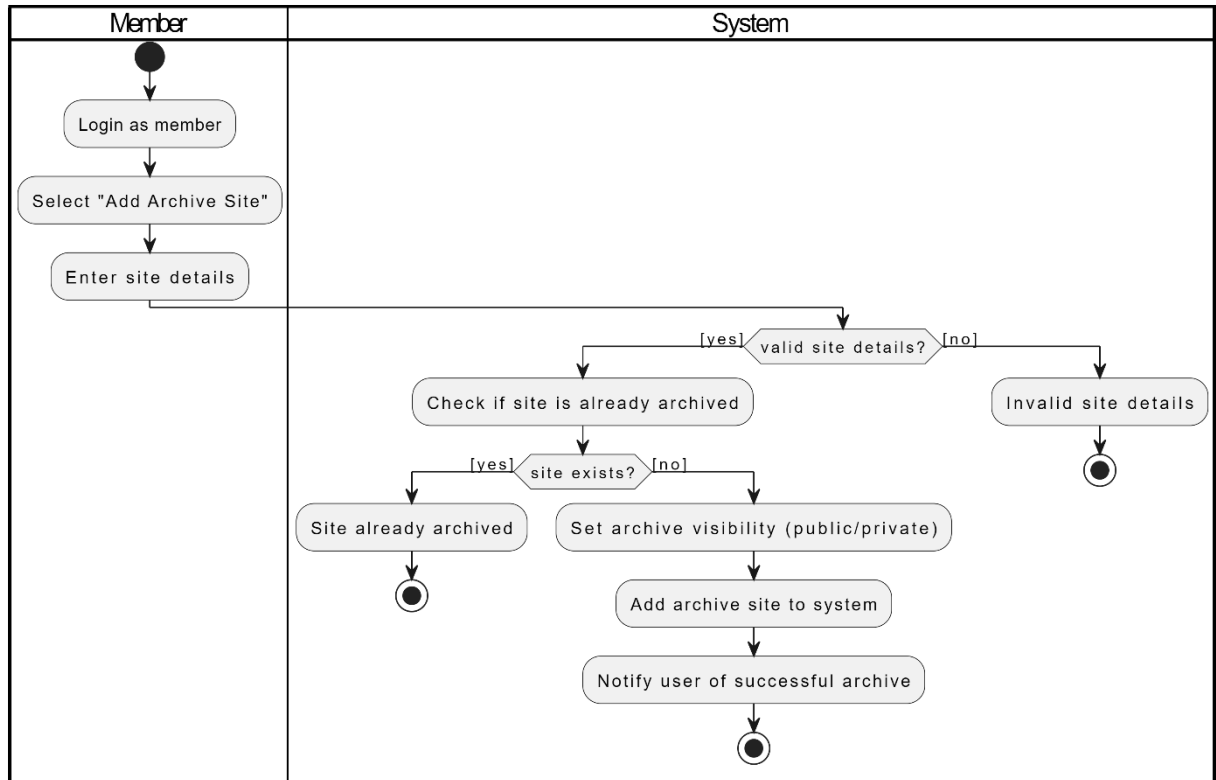


Figure 2.2-2 Swimlane diagram for adding a site for archiving

## 2.3 Sequence diagram

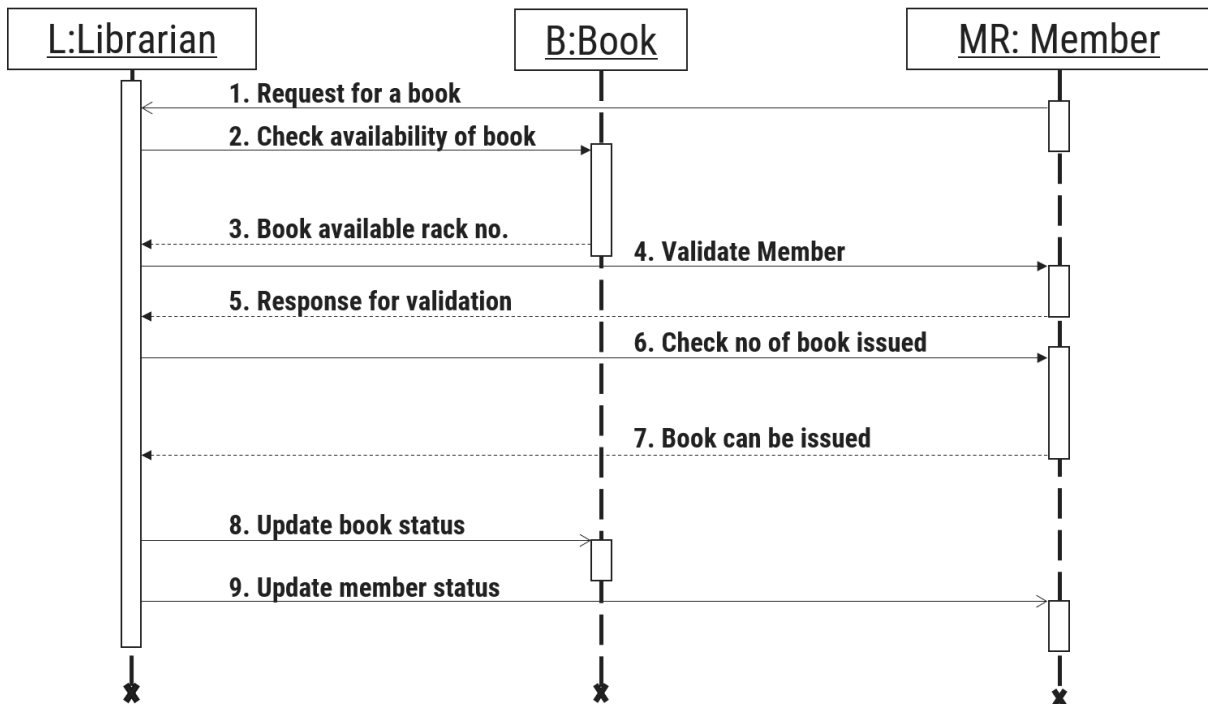


Figure 2.3-1 Sequence diagram for Book Issue

## 2.4 State diagram

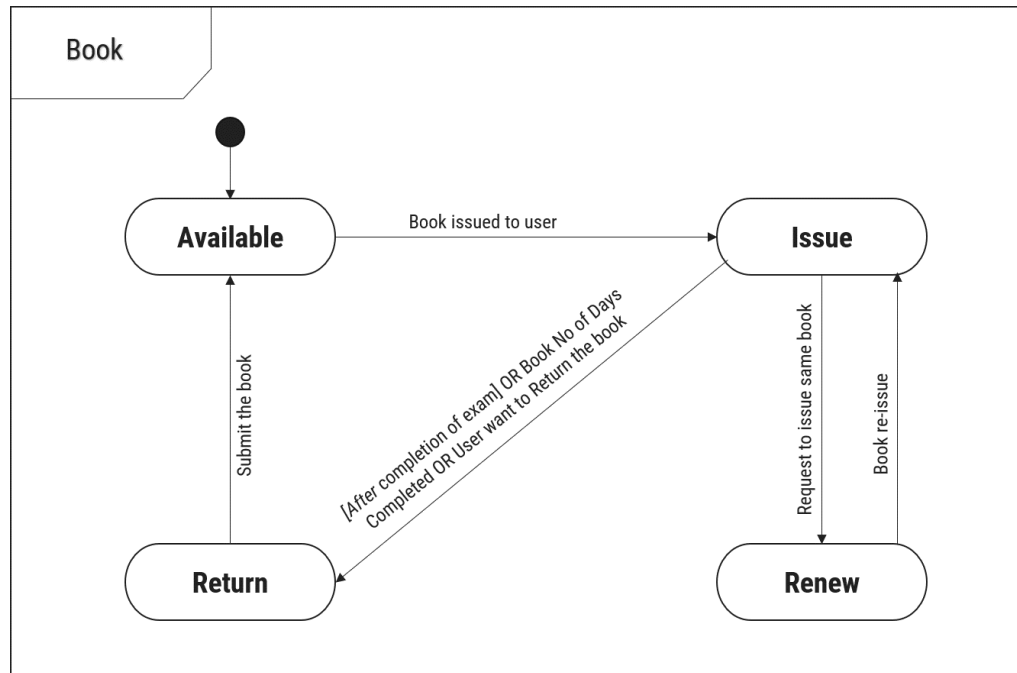


Figure 2.4-1 State diagram of Book

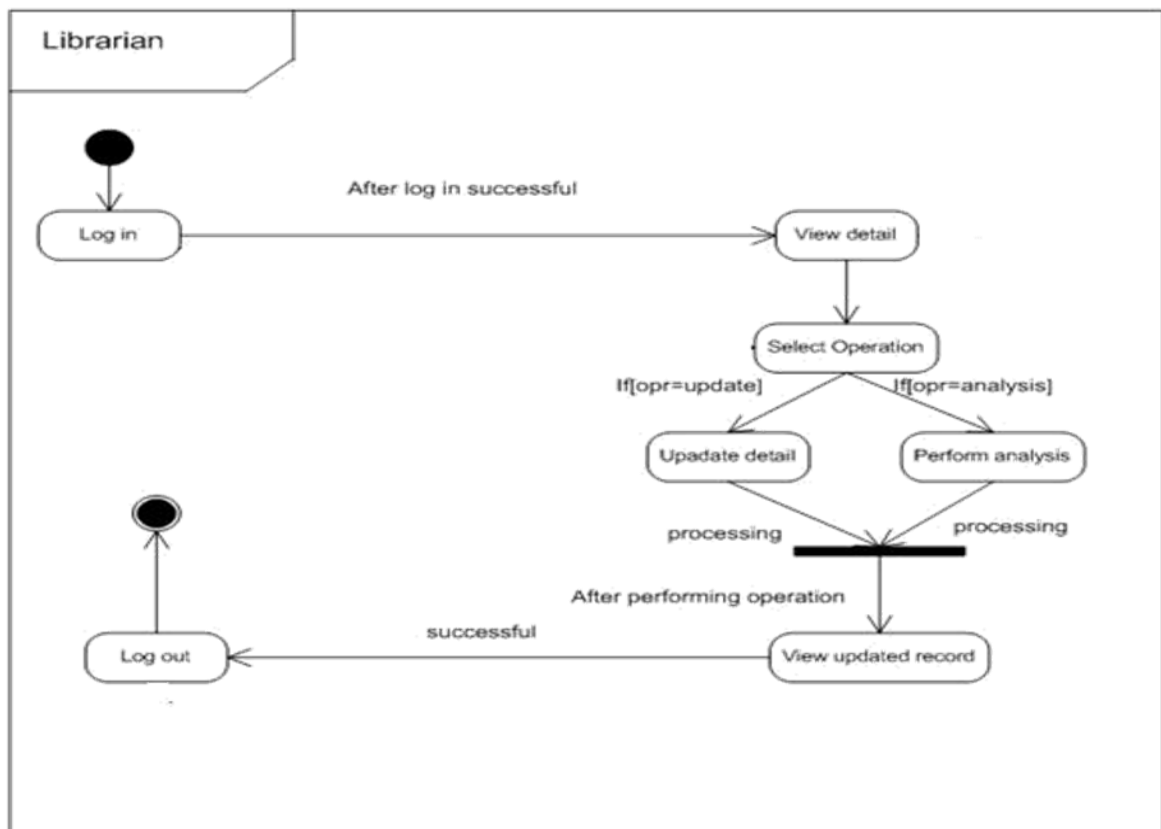


Figure 2.4-2 State diagram for Librarian

## 2.5 Class diagram

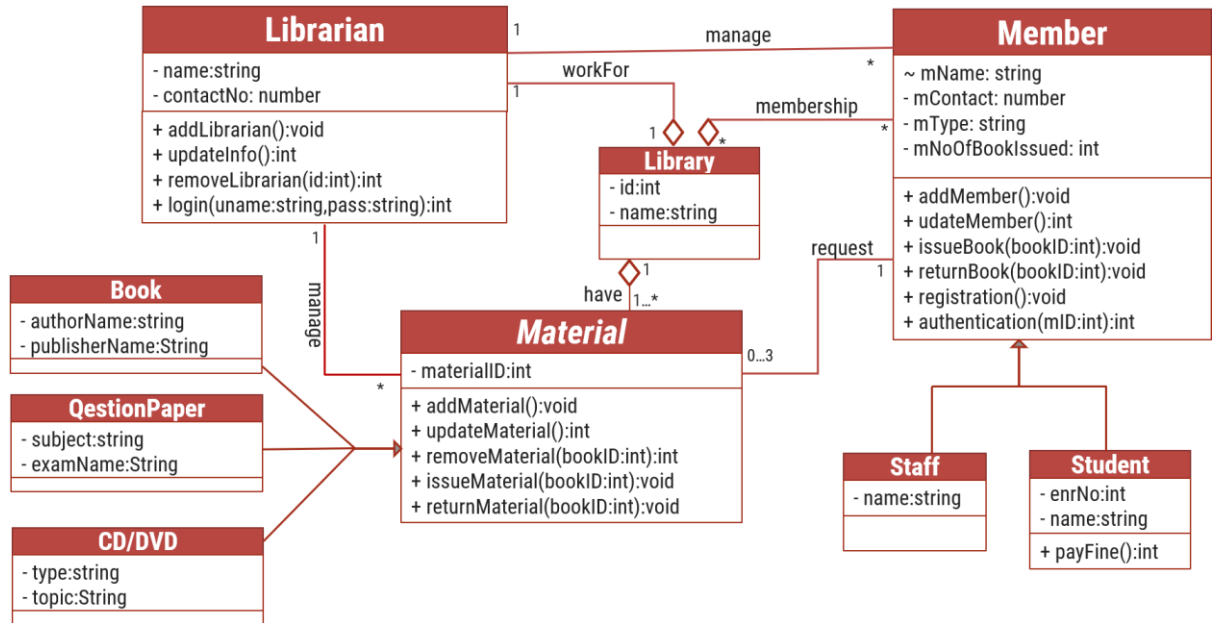


Figure 2.5-1 Class diagram for Library management system

## 2.6 Data flow diagram

### 2.6.1 Context diagram (level-0)

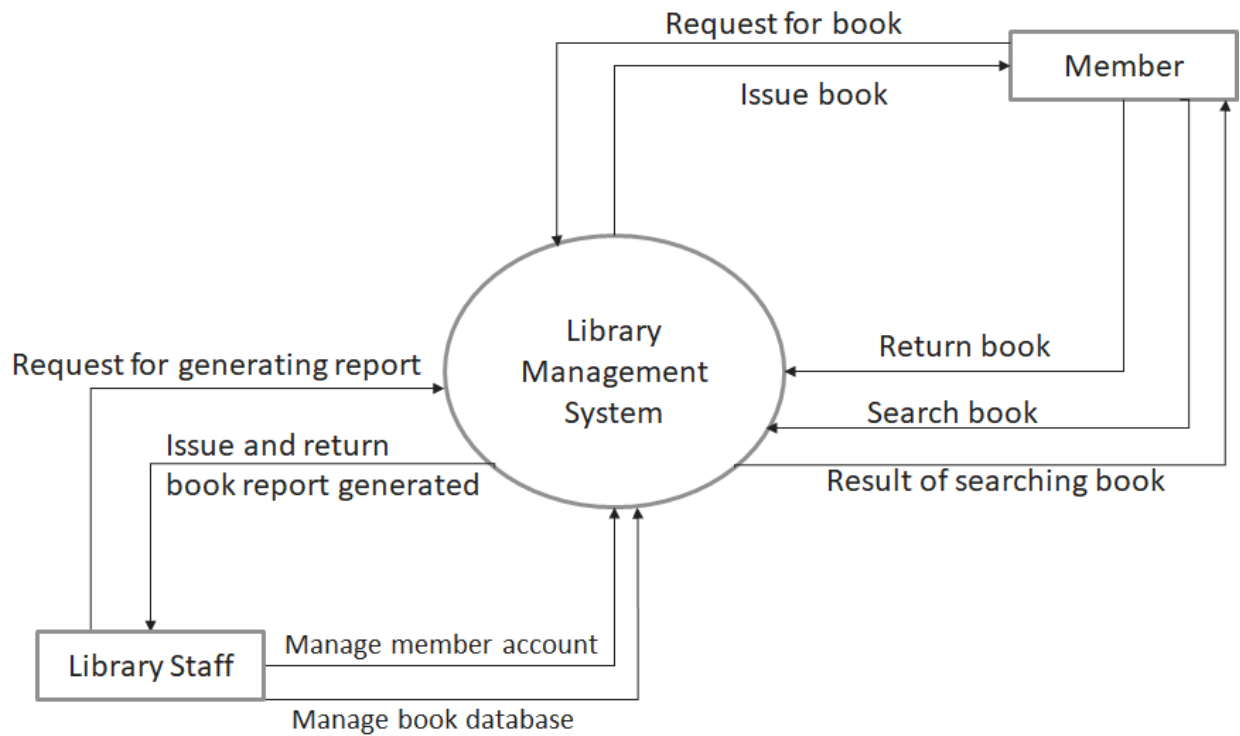


Figure 2.6-1 Context diagram for Library management system

### 2.6.2 DFD Level-1

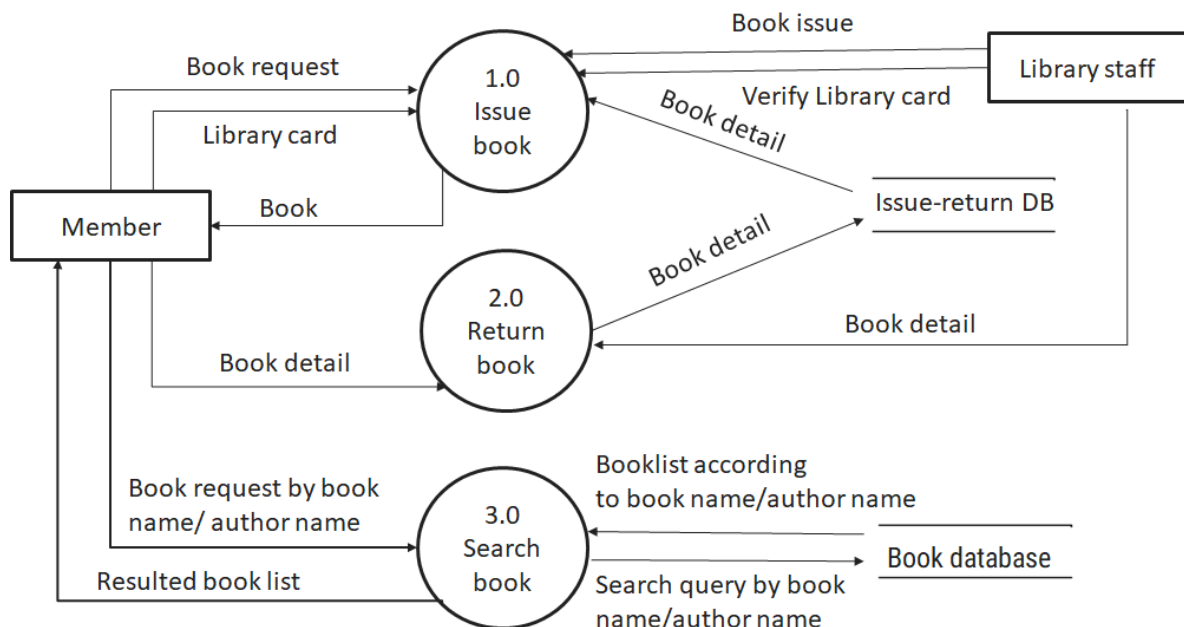


Figure 2.6-2 DFD level-1 for Library management system

## 2.6.3 DFD Level-2

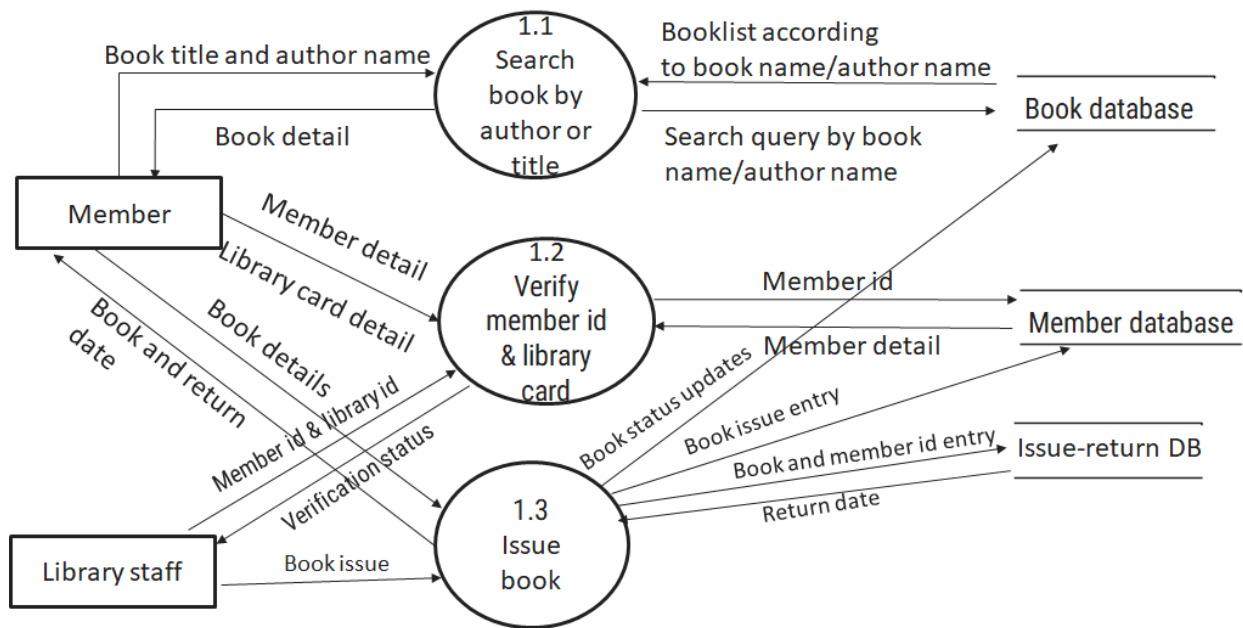


Figure 2.6-3 DFD level-2 for Issue book

### 3 External interface requirement (Screens)

#### 3.1 Screen-1: Registration Form

Figure 3.1-1 Screen-1: Registration Form

**Purpose:** This form will allow the target end-users to register in the system. To register , the following information will be encoded in the system.

Table 3.1-1 Screen element of Registration form

| Sr. | Screen Element              | Input Type | O/M   | 1/N   | Description                                                                                  |
|-----|-----------------------------|------------|-------|-------|----------------------------------------------------------------------------------------------|
| 1   | Username                    | Textbox    | M     | 1     | Username field should be editable and accept the Username.                                   |
| 2   | Email                       | Textbox    | M     | 1     | Email field should be editable and accept the email with proper format.                      |
| 3   | Password                    | Password   | M     | 1     | Password field should be editable and accept the password and display as star or dot.        |
| 4   | Retype password             | Password   | M     | 1     | Retype Password field should be editable and accept the password and display as star or dot. |
| 5   | I agree to the terms        | Checkbox   | M     | 1     | Checkbox for accepting terms and condition.                                                  |
| 6   | I already have a membership | Link       | ----- | ----- | Link for navigate to login page for registered user.                                         |
| 7   | Register                    | Button     | ----- | ----- | Register is a button for store the entered data into database.                               |

### 3.2 Screen-2: Login Form

Library System

Sign in to start your session

Username 

Password 

☐ **Remember Me** 

[I forgot my password](#)

[Register a new membership](#)

Figure 3.2-1 Screen-2: Login Form

**Purpose:** This form will be used by the system's users to access records and features of the system. The users will input the correct combination of their username and password to be able to login to the system.

Table 3.2-1 Screen element of Login form

| Sr. | Screen Element            | Input Type | O/M   | 1/N   | Description                                                                           |
|-----|---------------------------|------------|-------|-------|---------------------------------------------------------------------------------------|
| 1   | Username                  | Textbox    | M     | 1     | Username field should be editable and accept the Username.                            |
| 2   | Password                  | Password   | M     | 1     | Password field should be editable and accept the password and display as star or dot. |
| 3   | Remember Me               | Checkbox   | M     | 1     | Saving login credentials through remember me checkbox                                 |
| 4   | I forgot my password      | Link       | ----- | ----- | Link for navigate to Forgot password page for allows users to recover password.       |
| 5   | Register a new membership | Link       | ----- | ----- | Link for navigate to membership registration.                                         |
| 6   | Sign in                   | Button     | ----- | ----- | Login button navigates to another page even if valid login credentials.               |

### 3.3 Screen-3: Add borrower book

The screenshot shows a modal window titled "Add Borrower". It contains the following fields and controls:

- ID**: A text input field with a label "ID" above it.
- Student ID**: A text input field with a label "Student ID" above it.
- First Name**: A text input field with a label "First Name" above it.
- Middle Name**: A text input field with a label "Middle Name" above it.
- Last Name**: A text input field with a label "Last Name" above it.
- Contact**: A text input field with a label "Contact" above it.
- Category**: A text input field with a label "Category" above it.
- Grade ID**: A text input field with a label "Grade ID" above it.
- Buttons**: "Submit" (blue) and "Cancel" (grey) buttons at the bottom.

Figure 3.3-1 Screen-3: Add borrower book

**Purpose:** This module will allow the system administrator to add, edit, update or delete borrowers of book. The admin can add borrower information and manage it.

Table 3.3-1 Screen element of Add borrower book

| Sr. | Screen Element | Input Type | O/M   | 1/N   | Description                                                                            |
|-----|----------------|------------|-------|-------|----------------------------------------------------------------------------------------|
| 1   | Borrower ID    | Textbox    | M     | 1     | Borrower ID field should be editable and fill autogenerated Borrower ID.               |
| 2   | Student ID     | Textbox    | M     | 1     | Student ID field should be editable and accept Student barcode id.                     |
| 3   | First Name     | Textbox    | M     | 1     | First Name field should be editable and accept only string as a borrower first name.   |
| 4   | Middle Name    | Textbox    | O     | 1     | Middle Name field should be editable and accept only string as a borrower middle name. |
| 5   | Last Name      | Textbox    | M     | 1     | Last Name field should be editable and accept only string as a borrower last name.     |
| 6   | Contact        | Textbox    | M     | 1     | Contact field should be editable and accept only 10-digit mobile number.               |
| 7   | Grade Id       | Textbox    | O     | 1     | Auto fetch detail from the database if proper student id is entered.                   |
| 8   | Submit         | Button     | ----- | ----- | Submit is a button for store the entered data into database.                           |
| 9   | Cancel         | Button     | ----- | ----- | Cancel is a button for close add borrower dialog.                                      |

## 4 Database design

### 4.1 List of Tables

- Book
- Borrower
- Student
- Staff

Table 4.1-1 Table: Book

| Column           | Data Type    | Null | Keys & Constrains   | Default Value & Description |
|------------------|--------------|------|---------------------|-----------------------------|
| BookID           | int          | NN   | PK (Auto Increment) |                             |
| BookName         | varchar(100) | NN   |                     |                             |
| ISBN             | varchar(100) | AN   |                     |                             |
| Publication_year | int          | AN   |                     |                             |
| Language         | varchar(50)  | AN   |                     |                             |

Table 4.1-2 Table: Borrower

| Column           | Data Type    | Null | Keys & Constrains   | Default Value & Description          |
|------------------|--------------|------|---------------------|--------------------------------------|
| BorrowerID       | int          | NN   | PK (Auto Increment) |                                      |
| BookID           | varchar(100) | NN   | FK                  | Reference of Book Table              |
| BorrowedFromDate | DateTime     | AN   |                     |                                      |
| BorrowedToDate   | DateTime     | AN   |                     |                                      |
| ActualReturnDate | DateTime     | AN   |                     |                                      |
| IssuedBy         | int          | NN   | FK                  | Reference of Student and Staff Table |

Table 4.1-3 Table: Staff

| Column      | Data Type    | Null | Keys & Constrains   | Default Value & Description |
|-------------|--------------|------|---------------------|-----------------------------|
| StaffID     | int          | NN   | PK (Auto Increment) |                             |
| StaffName   | varchar(100) | AN   |                     |                             |
| IsAdmin     | Boolean      | AN   |                     |                             |
| Designation | varchar(100) | AN   |                     |                             |

Table 4.1-4 Table: Student

| Column      | Data Type    | Null | Keys & Constrains   | Default Value & Description |
|-------------|--------------|------|---------------------|-----------------------------|
| StudentID   | int          | NN   | PK (Auto Increment) |                             |
| StudentName | varchar(100) | AN   |                     |                             |
| Gender      | varchar(100) | AN   |                     |                             |
| DOB         | DateTime     | AN   |                     |                             |
| Department  | varchar(100) | AN   |                     |                             |
| Contact     | number(10,0) | AN   |                     |                             |

## 5 Stories and Scenario

### 5.1 Story-1: Add New Book in Library Catalogue

|                   |   |                                                                                                                                                     |
|-------------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Story # S1</b> | : | <b>As a</b> Librarian,<br><b>I want to</b> add a new book in library catalogue<br><b>So that</b> everyone can easily find and borrow it.            |
| <b>Priority</b>   | : | High                                                                                                                                                |
| <b>Estimate</b>   | : | XL                                                                                                                                                  |
| <b>Reason</b>     | : | The addition of a new book to the library catalogue is crucial for ensuring that the library's collection is up-to-date and accessible to everyone. |

#### 5.1.1 Scenario# S1.1

|                            |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scenario# S1.1</b>      | : | Adding a New Book with Valid Information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Prerequisite</b>        | : | Librarian is logged in to the Library management system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Acceptance Criteria</b> | : | <p><b>Given:</b> The Librarian is navigated to the library catalog management page. Valid book information, including title, author, ISBN, and other relevant details is added.</p> <p><b>When:</b><br/>The librarian selects the "Add New Book" option<br/>And The librarian enters valid book details<br/>The librarian clicks the "Save" button to add the book to the catalog.</p> <p><b>Then</b> the system successfully adds the book to the catalog and the librarian receives a confirmation message with the book's identification number.</p> |

#### 5.1.2 Scenario# S1.2

|                            |   |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scenario# S1.2</b>      | : | Adding a New Book with Invalid Information.                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Prerequisite</b>        | : | The librarian is logged into the library management system.                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Acceptance Criteria</b> | : | <p><b>Given:</b> The librarian is on the library catalogue management page</p> <p><b>When:</b> The librarian selects the "Add New Book" option and the librarian enters an incomplete or incorrect book details and librarian clicks the "Save" button to add the book to the catalogue.</p> <p><b>Then</b> the system displays error messages for the incorrect or missing information and the book is not added to the catalogue.</p> |

#### 5.1.3 Scenario# S1.3

|                            |   |                                                                                                                                                                                                                                                                                                                  |
|----------------------------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scenario# S1.3</b>      | : | Attempting to Add a Duplicate Book                                                                                                                                                                                                                                                                               |
| <b>Prerequisite</b>        | : | The librarian is logged into the library management system and the librarian is on the library catalogue management page                                                                                                                                                                                         |
| <b>Acceptance Criteria</b> | : | <p><b>Given:</b> The book information, including title, author, ISBN, and other relevant details, is available and the book with the same ISBN is already in the catalogue.</p> <p><b>When:</b> User Clicks on "Add book" button. Enter a number of copies with the same book detail mentioned in the field.</p> |

|  |                                                                                               |
|--|-----------------------------------------------------------------------------------------------|
|  | <b>Then:</b> Generate unique book id, barcode and spine label for various book of same title. |
|--|-----------------------------------------------------------------------------------------------|

## 5.2 Story-2: Search Book

|                   |                                                                                                                                                                                     |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Story # S2</b> | : <b>As a</b> Librarian or member,<br><b>I want to</b> search for books by title, author, or keyword,<br><b>So that</b> I can quickly find books that match my interests.           |
| <b>Priority</b>   | : High                                                                                                                                                                              |
| <b>Estimate</b>   | : M                                                                                                                                                                                 |
| <b>Reason</b>     | : Implementing a search functionality is essential for enhancing the user experience, as it allows librarian and member to efficiently discover and access the library's resources. |

## 5.3 Story-3: Manage due date for borrowed book

|                   |                                                                                                                                                                        |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Story # S3</b> | : <b>As</b> Librarian,<br><b>I want to</b> manage due dates for borrowed books<br><b>So that</b> I can ensure that books are returned on time and avoid overdue fines. |
| <b>Priority</b>   | : High                                                                                                                                                                 |
| <b>Estimate</b>   | : M                                                                                                                                                                    |
| <b>Reason</b>     | : Proper due date management is crucial for maintaining the library's collection and ensuring that books are available for all members.                                |

## 5.4 Story-4: Renew book

|                   |                                                                                                                                                           |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Story # S3</b> | : <b>As</b> Librarian,<br><b>I want to</b> renew a book that I have borrowed<br><b>So that</b> I can extend my borrowing period if needed.                |
| <b>Priority</b>   | : Medium                                                                                                                                                  |
| <b>Estimate</b>   | : M                                                                                                                                                       |
| <b>Reason</b>     | : Book renewal functionality is a convenience feature for librarian, allowing them to keep a book for an extended period if no one else has requested it. |

## 5.5 Story-5: Generate a report on book usage and availability

|                   |                                                                                                                                                                           |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Story # S3</b> | : <b>As</b> Librarian,<br><b>I want to</b> generate reports on library usage and book availability<br><b>So that</b> I can make informed decisions about library's books. |
| <b>Priority</b>   | : Medium                                                                                                                                                                  |
| <b>Estimate</b>   | : L                                                                                                                                                                       |
| <b>Reason</b>     | : Reporting functionality helps librarians track the usage of library resources and make data-driven decisions to improve services and collections.                       |

## 6 Test cases

|                         |                       |                             |                        |
|-------------------------|-----------------------|-----------------------------|------------------------|
| <b>Project Name:</b>    | <b>EMI Calculator</b> | <b>Test Designed by:</b>    | <b>P. U. Jadeja</b>    |
| <b>Module Name:</b>     | <b>Login</b>          | <b>Test Designed date:</b>  | <b>01-10-2023</b>      |
| <b>Release Version:</b> | <b>1.0</b>            | <b>Test Executed by:</b>    | <b>R. B. Gondaliya</b> |
|                         |                       | <b>Test Execution date:</b> | <b>15-01-2023</b>      |

| Pre-condition: Web application should be accessible |                                                  |            |                                                                               |              |
|-----------------------------------------------------|--------------------------------------------------|------------|-------------------------------------------------------------------------------|--------------|
| Test Case ID                                        | Test Title                                       | Test Type  | Description                                                                   | Test Case ID |
| TC_001                                              | Login to web application with valid credential   | Functional | Login to Library management system web application through valid credential   | TC_001       |
| TC_002                                              | Login to web application with invalid credential | Functional | Login to Library management system web application through invalid credential | TC_002       |
| TC_003                                              | Varify login page elements                       | GUI        | varify that all elements are available on login page                          | TC_003       |

|                        |                                                |
|------------------------|------------------------------------------------|
| <b>Test Case Title</b> | Login to web application with valid credential |
| <b>Test Type</b>       | Functional                                     |
| <b>Test Priority</b>   | High                                           |
| <b>Pre-condition</b>   | Web application should be accessible           |

| Test Step | Test Case Description                    | Expected Result                                                                      | Actual Result                                | Status | Comment                                     | Data                                                                                            | BUG ID |
|-----------|------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------|--------|---------------------------------------------|-------------------------------------------------------------------------------------------------|--------|
| 1         | Access Web application URL               | The site launched properly                                                           | Site launched successfully                   | Pass   |                                             | <a href="https://accounts.google.com/ServiceLogin">https://accounts.google.com/ServiceLogin</a> |        |
| 2         | Enter valid Username in username field   | Username field should be editable and accept the Username                            | Username input accepted                      | Pass   |                                             | Username: Rbgondaliya@gmail.com                                                                 |        |
| 3         | Enter valid Password in Password field   | Password field should be editable and accept the password and display as star or dot | Password input displayed in dot and accepted | pass   |                                             | Password: rbgondaliya                                                                           |        |
| 4         | Enter valid captcha code in captch field | Captch field should be editable and accept captcha and captcha is case sensitive     | Captcha input accepted                       | Pass   | Step require d when human action validation | get captcha from image which is near by captcha field                                           |        |

|   |                       |                                                        |                                                                                      |      |         |  |  |
|---|-----------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------|------|---------|--|--|
|   |                       |                                                        |                                                                                      |      | perform |  |  |
| 5 | Click on login button | User should login into site and navigated to dashboard | User navigated to dashboard and username should be display in top of the right side. | pass |         |  |  |

|                 |                                                  |
|-----------------|--------------------------------------------------|
| Test Case Title | Login to web application with invalid credential |
| Test Type       | Functional                                       |
| Test Priority   | Medium                                           |
| Pre-condition   | Web application should be accessible             |

| Test Step | Test Case Description                                                            | Expected Result                                                     | Actual Result                                   | Status | Comment | Data | Bug ID |
|-----------|----------------------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------|--------|---------|------|--------|
| 1         | Verify that User is not able to Login with invalid Username and invalid Password | Should be display an error message enter wrong username or password | Display an error of wrong username and password | Pass   |         |      |        |
| 2         | Verify that User is not able to Login with Valid Username and invalid Password   | Should be display an error message enter wrong password             | Display an error of wrong password              | Pass   |         |      |        |
| 3         | Verify that User is not able to Login with                                       | Should be display an error message                                  | Display an error Username not found             | Pass   |         |      |        |

|   |                                                                       |                                                                 |                                                 |      |                                          |  |         |
|---|-----------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------|------|------------------------------------------|--|---------|
|   | invalid Username and Valid Password                                   | User not found                                                  |                                                 |      |                                          |  |         |
| 4 | Verify that User is not able to Login with blank Username or Password | Set required field validation message for Username and Password | Display an error of wrong username and password | Fail | Not perform a validation function fix it |  | Bug_002 |

|                 |                                      |
|-----------------|--------------------------------------|
| Test Case Title | Varify login page elements           |
| Test Type       | GUI                                  |
| Test Priority   | Medium                               |
| Pre-condition   | Web application should be accessible |

| Test Step | Test Case Description                                                                                                                                                     | Expected Result                                   | Actual Result                  | Status | Comment | Data                                                                                            | Bug ID |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|--------------------------------|--------|---------|-------------------------------------------------------------------------------------------------|--------|
| 1         | Launch application with the given url                                                                                                                                     | The site launched properly                        | Site launched successfully     | Pass   |         | <a href="https://accounts.google.com/ServiceLogin">https://accounts.google.com/ServiceLogin</a> |        |
| 2         | Verify that the login screen contains elements such as Username, Password, Sign in button, Remember password check box, Forgot password link, and Create an account link. | All listed control displayed properly on the page | Login page loaded successfully | Pass   |         |                                                                                                 |        |

|   |                                                                                                           |                                                                     |                                                                   |      |                                               |  |  |
|---|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------|------|-----------------------------------------------|--|--|
| 3 | Verify that cursor is focused on "Username" text box on the page load                                     | Cursor is focused in Username textbox                               | Cursor focus in Username textbox                                  | Pass |                                               |  |  |
| 4 | Verify that tab functionality is working properly or not                                                  | When tab pressed cursor move in next control                        | Cursor moving in next control                                     | Pass |                                               |  |  |
| 5 | Verify that all the fields such as Username, Password has a valid placeholder                             | All text fields have proper placeholder                             | All text fields have proper placeholder                           | Pass |                                               |  |  |
| 6 | Verify that the labels float upward when the text field is in focus or filled (In case of floating label) | When field is focused or filled, label display on top of the filled | When field is focus or filled, label display on top of the filled | Pass | step required when fields with floating label |  |  |
| 7 | verify that forgot password link working properly                                                         | when click on forgot password load forgot password page             | forgot password link not working                                  | Fail |                                               |  |  |

## 7 References

- [http://www.w3schools.com/html/html\\_intro.asp](http://www.w3schools.com/html/html_intro.asp)
- <https://www.w3schools.com/php/default.asp>
- <https://www.javatpoint.com/uml>